

Energy-Aware IoT based solution for smart Beehives

1st Nourhene Ellouze

LIPAH research Lab

Faculty of Sciences of Tunis
University of Tunis El Manar
norhene.ellouze@fst.utm.tn

2nd Cyrine Bouallegue

LIPAH research Lab

Faculty of Sciences of Tunis
University of Tunis El Manar
syrine.bouallegue@etudiant-fst.utm.tn

3rd Hella Kaffel Ben Ayed

LIPAH research Lab

Faculty of Sciences of Tunis
University of Tunis El Manar
hella.kaffel@fst.utm.tn

Abstract—Beekeeping is vital for agriculture and environmental sustainability, but traditional hive management methods are often time-consuming. IoT-based smart beehive systems provide a modern solution, offering monitoring and improved hive management. However, these systems face significant challenges, such as high energy consumption, data reliability issues, and limited communication efficiency, particularly in remote and rural areas. This paper proposes an energy-efficient IoT-based beehive solution designed to cope with these challenges. This solution is based on careful selection of parameters monitoring bees and energy-efficient communication protocols. It also implements a set of data sampling and transmission heuristics to reduce energy consumption while ensuring reliable data collection and transmission. Furthermore, the proposed architecture promotes an adaptable solution that supports continuous monitoring in diverse environments, depending on whether the beehives are in areas with communication coverage, or rural areas without such coverage. Simulation results confirm the system's ability to achieve energy efficiency without compromising monitoring quality, making it a sustainable solution for beekeeping.

Index Terms—Beekeeping, monitoring bees health status, smart beehives, IoT, energy preservation, heuristic algorithms.

I. INTRODUCTION

Beekeeping has been an important part of agriculture and natural ecology, helping with pollination, food production, and keeping ecosystems balanced. To this end, it is crucial to ensure a regular inspection of beehives to ensure the monitoring and the sustaining of the health of bee populations. Traditional beekeeping is based on manual inspection and physical examination of beehives. However, these methods exhibit several limitations. First, they are susceptible to human errors, especially if the examination is performed by novice beekeepers. Second, manual inspection can upset the balance of bees leading to a decrease in productivity. Third, these methods could not promote accurate verification of hive conditions. Fourth, manual inspection can be laborious and costly, especially if the hives are placed over large rural regions.

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To cope with these limitations, a honeybee monitoring system should be designed. It should: a) rely on the use of hives equipped with sensors to monitor environmental conditions and the health and activities of bees; b) be scalable in terms of the number of sensors and the sampling rate of data; c) integrate low-cost, reusable, and energy-efficient equipments; and d) ensure the exchange and storage of the collected data.

Several research works addressing bees monitoring have been proposed in the literature. Some of these works [1]–[5] have addressed only the design of the monitoring system. Other works [4], [6]–[8] proposed a limited number of times to collect and transmit data, to preserve energy. However, they exhibit the absence of an end-to-end architecture detailing the components and protocols involved from the collection of data generated by sensors, to the storage. They also lack effective methods ensuring energy preservation.

In this paper, we propose an energy-efficient Internet of Things (IoT) based smart beehive solution allowing to cope with challenges mentioned previously. Our contributions are fourfold. First, the proposed architecture integrates IoT equipment that have been carefully selected to promote efficient bee's health surveillance, low-cost design, and energy-aware solution. Second, the architecture integrates energy-efficient communication protocols that are selected based on a set of criteria, including low energy consumption, communication range, and absence of impact on the bees health. Third, the proposed architecture is customizable to two scenarios, depending on whether the beehives are located in areas with public communication coverage or remote rural areas without such coverage. Fourth, a set of sampling and data transmission heuristics are proposed to reduce energy consumption while maintaining reliable data collection and transmission.

The remainder of the paper is organized as follows. The next section provides a literature review of IoT solutions for beekeeping. In section III, we describe the requirements and data parameters required to build an efficient beehive monitoring system. Section IV details the proposed architecture. Section V describes the data sampling and transmission heuristics. In section VI, we present the simulation model and evaluate the proposed data sampling heuristics. Section VII concludes the paper and presents the future work.

II. OVERVIEW OF IoT SOLUTIONS FOR BEEKEEPING

In this section, we review IoT based beekeeping solutions.

To monitor the health status of bee colonies and surrounding weather conditions, some research works proposed the design of standalone smart hives, which collect and transmit data to remote servers. In [1], the authors proposed an IoT-based monitoring system for hives, which integrates hardware components to collect and transmit data from hives; and software components to promote real-time monitoring and notification management. The effectiveness and real deployment of this system were shown in [2]. The authors of [3] designed a system that combines hardware and software components. Various sensors are placed inside and outside the hive to monitor different parameters. The software components include: a computational module to perform analytical and predictive assessment of sensory data; and a web application to visualize data. All of these work [1]–[3] highlight the potential of IoT in improving beekeeping practices through enhanced monitoring and data-driven decision-making. However, they only focused on the development of software and hardware parts without addressing the communication protocols to ensure data transmission. They also lack the proposal of efficient methods to preserve energy and deal with frequent batteries replacement.

Other IoT solutions for beekeeping integrate the use of an intermediate node to collect data from hives using short-range communication protocols, then transfer data to remote servers using cellular networks. In [4], the proposed system includes a set of smart hives, where each is equipped with a detection device that collects measurements using sensors and sends them to a router using WiFi. The router, which acts as a gateway, creating a Wi-Fi hotspot from mobile networks (2G/3G), sends the received data to the cloud server. Another system, proposed in [5], includes five monitoring nodes (hives) that are powered by batteries and one main node that is powered by a solar power station. Every 30 minutes (min), each monitoring node collects data and transmits them to the main node using the ESP-NOW protocol, which is an energy-efficient data exchange protocol. The main node, which integrates an ESP32 microchip and a GSM module, transmits the received data to a remote server. The systems proposed in [4], [5] promote efficient bees monitoring. However, they exhibit a high dependence on cellular networks, which could not always be available, especially in rural areas. In addition, they do not address the challenges of energy consumption.

To reduce energy consumption, some work developing IoT based solutions for bees monitoring [4], [6]–[8] proposed predefined periods of times to sample data parameters and fixed numbers of times to send collected sensory data. A summary of these proposals is illustrated in Table I. These proposals lack energy-efficient methods to reduce energy consumption while ensuring efficient monitoring of bees health.

III. REQUIREMENTS AND DATA SPECIFICATIONS FOR EFFICIENT BEEHIVE MONITORING

The identification of system requirements and collected data parameters is crucial for the success of the system design.

A. System requirements

To address the limitations discussed above, a beehive monitoring system should meet the following requirements.

Scalability: The system should ensure the monitoring of various parameters indicating environmental conditions, bees health status, and bees productivity. It should also enable the incorporation of additional sensors to monitor new parameters.

Reduced Cost: To guarantee that the system costs remain within the expected budget of beekeepers, the benefits of the system should be carefully addressed. To this end, the system should integrate low-cost devices that do not affect the system's performance. The use of inexpensive and reusable IoT equipment makes cost-effectiveness possible. The selection of the most critical sensors can also reduce costs while providing the necessary data for hives monitoring, which is particularly important during the transhumance period, for example.

Energy minimization: Energy optimization is a crucial consideration in the system to reduce the frequency of batteries replacement. Especially smart beehives can be placed in rural areas, where batteries cannot be replaced frequently. The system components must be carefully selected to avoid using energy-intensive components. They should also follow sleep and waking schedules, to reduce energy consumption. It is also important to reduce the energy overhead induced by data processing and transmission to the maximum possible.

Data Accessibility: The system should guarantee the access, collection, and storage of data gathered by smart beehives independently of the presence of coverage of the public network. In fact, beehives could be placed in rural regions, where public network coverage is not available.

B. Essential Data Parameters for Smart Beehive Monitoring

The objectives of a smart beehive are monitoring the health status of bees, the environmental conditions, and the bees productivity. Based on these objectives and research work [3], [4], [9], we identify the types of data to be collected.

Temperature: It is crucial to monitor the temperature level inside and outside the hives (internal and ambient temperature, respectively). The ambient temperature level allows identifying the environmental conditions. The internal temperature level is an indicator of the health status of the bee colony. It allows to detect and even predict the occurrence of crucial events (e.g.; swarming, weakening, or even the bees death). In particular, the excessive decrease of the internal temperature (e.g.; below 20 °C) can induce the death of bees [10]. A significant increase of internal temperature (e.g.; higher than 35 °C) can also induce the bees migration from hives.

Humidity: Humidity inside the hive highly impacts the development of bee brood (eggs, larvae, and pupae), the bees productivity, and the appearance of parasites and pathogens.

Sounds and Vibrations: The analysis of sound and vibration patterns can provide information related to the health and activities of a bee colony [11]. Their signatures allow the detection of critical events (e.g.; swarming, queen loss).

TABLE I
COMPARATIVE TABLE OF DATA SAMPLING AND TRANSMISSION FREQUENCIES

	Sampling Frequencies						Transmission	
	Temperature	Movement	Humidity	Sound	Gas	Weight	Frequencies	Methods
[6]	6 times per day	-	6 times per day	-	6 times per day	-	at 12:00 every day	Zigbee/3G/GSM
[7]	6 times per day	-	6 times per day	-	3 times per day	-	3 to 6 times per day	ZigBee/3G
[8]	Every 15mn	Every 15 mn	-	-	-	Every 5 mn	Every 5 or 15 mn	GSM/4G/Wi-Fi
[4]	Every hour	-	Every hour	Every hour	-	Every hour	Every hour	Wi-Fi/3G

Gas: Monitoring gas (such as carbon dioxide - CO₂) allows early detection of ventilation issues and prevention of physiological stress-related issues.

Weight: The analysis of the variation of the hive weight can provide various information, including bee population, amount of honey stored, volume of pollen, nectar collected and food consumption. Mathematical equations, proposed in [12], allows the estimation of factors related to the foraging and food processing activities of bees based on weight measurements. The weight decrease may also indicate theft.

Rainfall: Identifying rainfall periods can be helpful. In [13], it was shown that extreme weather events causes a set of challenges related to productivity. Among as, the capacity of bees to forage can be limited by rain. The availability of food may also decrease, as nectar sources become harder to reach.

Data collection in hives involves measuring both internal and external parameters to ensure comprehensive monitoring of environmental conditions, bees health, and productivity.

IV. ARCHITECTURE AND SYSTEM DESIGN

This section details the architecture of the proposed solution.

A. System Architecture

The system architecture is customizable to two scenarios, depending on whether the beehives are located in areas with public communication coverage or rural areas without such coverage. The system components are detailed in the sequel.

Smart Beehives: are placed close to each other around the gateway within a predefined range. Each hive is equipped with a set of sensors to monitor parameters related to the environmental conditions and the bees health status, and a micro-controller to process the collected data.

Gateway: is placed in the proximity of beehives, within a range of several meters. It performs the following functions: 1) the collection of sensory data sent from smart beehives within its range; 2) the storage of the collected data in its internal memory for a predefined period; and 3) the transmission of the collected data to a remote server (e.g.; cloud or private server). To this end, the gateway is equipped with an antenna providing a short-range communication protocol to receive data from beehives. It is also equipped with dual communication capabilities to handle different network scenarios:

- **With Public Network Coverage** (Figure 1): the gateway uses cellular networks to transmit the collected data to the remote server. This enables the remote monitoring of hives.

- **Without Public Network Coverage** (Figure 2): the gateway acts as a local data hub. Beekeepers can collect data from it using their smartphones via Bluetooth or Wi-Fi.

Data Storage: With public network connectivity, data are stored in a remote storage system. This setup supports automated analysis and historical storage of data, allowing beekeepers to monitor trends and receive alerts on potential issues. The absence of network coverage induces the storage of data in the gateway's internal memory. Upon re-establishing a connection to a public network, the gateway transmits the stored data and then deletes it from its internal memory.

Mobile and Web applications: enable the visualization of data, and the notifications of unusual conditions or issues.

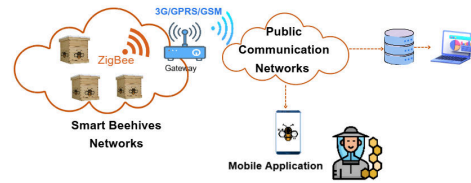


Fig. 1. System architecture: presence of public communication coverage



Fig. 2. System architecture: absence of public communication coverage

B. Communication Protocols

The selection of communication protocols is one of the crucial requirements to promote energy-aware solution. In fact, the energy consumption induced by the transmission of data presents a significant proportion of the total energy consumed by the system. Two categories of communication protocols are integrated into our architecture. The first category includes short-range communication protocols that ensure communication between smart hives and the gateway and between the gateway and the beekeepers smartphone. The second category is long-range communication protocols to transmit data collected by the gateway through public communication networks to be stored in remote storage system. Both protocol categories should be selected according to the following detailed criteria.

1) *Selection criteria of communication protocols:* The implemented protocols should consider the following criteria:

Energy Efficiency: It is crucial to select communication protocols that induce low-power energy consumption to prevent the frequent replacement of batteries. This makes the system more durable and practical for long-term use.

Absence of harmful impacts on the health of bees: Bees are sensitive to specific frequencies of electromagnetic waves induced by the antennas used to transmit data. The increasing presence of wireless technology can affect the bees health and behavior. Studies in [14], [15] showed that radio waves at frequencies 900 MHz, 2.45 GHz, and 5.805 GHz can affect bees health. When a mobile phone emitting 900 MHz radiation was placed in a hive, young queen bees exposed to it had a lower chance of hatching. Long-term exposure to 2.45 GHz and 5.805 GHz can confuse foraging bees, making it harder for them to return to the hive. While some bees recover, too much exposure to wireless signals could harm bee populations.

2) *The implemented communication protocols:* They are selected on the basis of the selection criteria discussed above.

Zigbee for short-Range communication: Zigbee is selected to transmit data from smart beehives to the gateway. As discussed in [16], Zigbee is an energy-effective protocol for data transmission in short-range distances. It is particularly well-suited for wireless sensor networks where low power consumption is critical to extend battery lifetime. In addition, Zigbee induces minimal electromagnetic emissions, which reduces the risk of interfering with bees natural behavior, making it a safe choice for use within hives.

3G/GPRS/GSM for long-Range communication: 3G/GPRS/GSM is selected to transmit data from the gateway to the remote server through public communication networks. It is selected for its reliability and ability to operate at frequencies that do not disrupt the natural routines of bees.

V. MANAGING DATA SAMPLING AND TRANSMISSION FOR REDUCING ENERGY CONSUMPTION

In this section, we propose a set of heuristics to minimize the power consumption of the system while ensuring reliable monitoring of hives. These heuristics allow beekeepers to select the approach that aligns with their operational needs.

A. Data Sampling Heuristics

To reduce the energy consumption, we propose heuristics allowing the management of the data sampling frequencies.

Heuristic 1 (H_1) - Periodic sensor activation: The principle behind this heuristic is to periodically activate all sensors integrated within the hive. Rather than running continuously, sensors are activated at specific time intervals to collect data, then switch to a low-power standby mode in between. Focusing on scheduled updates ensures that the hive's conditions are monitored regularly without the power drain of continuous monitoring. Widely used in research, periodic activation provides a practical and energy-efficient solution when regular data collection is sufficient for effective monitoring.

Heuristic 2 (H_2) - Temperature threshold based activation: This heuristic relies on the temperature sensor to initiate additional data collection by the remaining sensors of the hive. Other sensors switch to active mode, only when the temperature sensor detects a predefined significant temperature variation (e.g., a shift exceeding 4°C) indicating a potential hive event that needs closer attention. When temperature remains stable, only the temperature sensor switches from standby to active mode every predefined period. H_2 allows reducing power usage while ensuring critical events detection.

Heuristic 3 (H_3) - Temperature interval based activation: The principal behind this heuristic consists in periodically activating the temperature sensor for sampling. If the temperature value is not within a predefined temperature interval (e.g., [20°C, 35°C]), other sensors are activated. Once data is collected, all sensors return to sleep mode. This heuristic helps detecting significant events without constant data collection, making it ideal for energy-constrained environment.

Heuristic 4 (H_4) - Seasonally adaptive period and threshold selection: This strategy adjusts how often sensors activate and what temperature thresholds they use based on the season. This helps save energy by matching data collection to expected weather changes. For example, in seasons like spring, sensors check more often with lower temperature thresholds. In winter, when conditions are more stable, sensors check less frequently and use higher thresholds, reducing unnecessary activities.

B. Data Transmission Heuristics

Data transmission present additional energy management challenges. To this end, we propose the following heuristics.

Periodic data transmission: The principle of this heuristic consists in sending data at a predefined time period, rather than the continuous data transmission. This guarantees preserving energy while maintaining regular updates on hive conditions. This also reduce the bees disruption induced by the antenna waves of the communication protocols.

Critical event detection based data transmission: The principle of this heuristic consists in sending data upon the detection of a critical event, where a critical event is a sampling value of a given parameter that is not within the normal range (e.g., temperature below 20 °C, CO2 higher than 4.3%). This heuristic allows minimizing energy consumption, while ensuring the early detection of critical events within the hives.

VI. ENERGY CONSUMPTION EVALUATION

The aim is to evaluate the data sampling heuristics. The simulation model and the results are presented in the sequel.

A. Simulation Model

We consider a single hive equipped with a set of sensors and a micro-controller. Datasets simulate data collected by sensors.

Data Sets: Two datasets that are collected under different environmental conditions and seasons are used. The first dataset [3], gathered in October, includes internal and external temperature and humidity, rain, air quality, etc. This period, marking the onset of winter, provide an unstable climate

ideal for testing the system in various conditions. The second dataset [5], collected during June, July, and August, focuses on temperature data collected from five hives. July, characterized by extreme and stable heat, allowed us to evaluate the system's efficiency under high-temperature conditions.

Sensor characteristics: The technical specifications of the sensors used in the simulation are detailed in Table II. The data types and resolution of each sensor are described as follows.

- Temperature and humidity: Digital output (16-bit).
- Sound: Binary (1-bit) or analog (10-bit) signal.
- Vibration: Binary output (1-bit).
- Weight: Digital output up to 24-bit resolution.
- Gas: Analog output with 12-bit resolution.

TABLE II
TECHNICAL SPECIFICATIONS OF SENSORS

Sensor Type	Specifications			
	Activation Time	Standby Power	Active Power	Operating Voltage
Temperature and Humidity [17]	2 sec	40 μ A	1 mA	3.3V
Weight [18]	2 sec	40 μ A	1.6 mA	5.5V
Sound [19]	2 sec	0.001 μ A	0.000150 A	5V
Gas [20]	3 sec	40 μ A	0.102 A	4.9V
Vibration [21]	2 sec	40 μ A	0.2 A	12V

Micro-controller: We use the STM32 Nucleo micro-controller for efficient processing and low power usage. It integrates 512 KB flash memory and uses various modes:

- **Active:** Full operation, all peripherals running (20,500 μ A).
- **Sleep:** CPU halted, peripherals active, low power (5.50 mA).
- **Stop:** CPU and clocks disabled, SRAM retained (732 μ A).
- **Standby:** Minimal power, RTC active, SRAM lost (3 μ A).

Estimating storage helps determine the micro-controller's capacity for extended data collection. With 512 KB of flash memory, the total data per cycle is expressed by Equation 1.

$$\text{Total data per cycle} = \sum_{i=1}^n D_i \quad (1)$$

where: a) Total data per cycle is the amount of data generated by sensors in one cycle (bytes); b) D_i is the data produced by sensor i in a cycle (bytes); and c) n is the number of sensors. Given that each cycle generates 79 bits, the number of stored cycles is equal to $\frac{4194304 \text{ bits}}{79 \text{ bits/cycle}} \approx 53,095$. Therefore, a 512KB flash memory could support more than 30 days storage of data collected using minute-level sampling executed by 6 sensors.

B. Evaluation results of data sampling heuristics

Due to the lack of useful datasets, we only simulate H_1 , H_2 and H_3 heuristics. The simulation results are detailed in the following.

Energy consumption using H_1 : Energy consumed by sensors using H_1 is evaluated using mathematical equations. The energy use per sensor is expressed as: $E = \text{Voltage} \times \text{Current} \times \text{Activation Time}$. Subsequently, the average energy use by sensors per day is expressed by Equation 2. Where:

- E_i : Energy consumed by an active sensor i (Joules).
- S_i : Energy consumed by a standby sensor i (Joules).
- n : Total number of sensors.

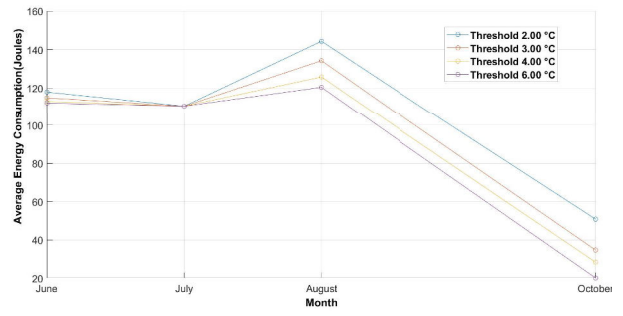


Fig. 3. Energy consumption per month using heuristic 2

- P : Period between 2 successive sensors activation (min).

$$\text{Average energy per day} = \frac{24 \times 60}{P} \cdot \sum_{i=1}^n (E_i + S_i) \quad (2)$$

By calculating the energy consumed by the micro-controller to perform its functions and the average energy of all sensors per day, the value of the average energy consumption per day is approximately equal to 274.08 Joules (where $P = 1$ hour).

Estimation of energy consumption using H_2 : The heuristic H_2 optimizes energy by activating sensors based on temperature thresholds ranging from 2°C to 6°C, with an activation period equal to 1 hour. As shown in Figure 3, low temperature thresholds (2°C) induce frequent sensor activations and increase energy use and storage needs, while higher thresholds (6°C) reduce activations and conserve energy. In July, energy use remains stable across all temperature thresholds as a result of the low temperature variations. Lower temperature thresholds improve monitoring accuracy but require more resources, whereas higher thresholds enhance energy efficiency.

Energy consumption using H_3 : This heuristic improves energy efficiency by activating sensors only when the temperatures go outside a specific range [20°C, 35°C] that could negatively impact the bees. As shown in Figure 4, energy use remains low when temperature is within this interval. In July, sensor activations are minimized as a result of stable temperature variation within the predefined interval, showing that this heuristic helps conserve energy and storage. However, during extreme temperature fluctuations, more data need to be collected, increasing resource use. This method balances accurate monitoring with efficient resource management.

Discussing the efficiency of data sampling heuristics: Figure 5 shows a comparative analysis of the first three heuristics, where H_1 is the widely used method in the literature. H_1 consumes twice the amount of energy compared to H_2 and H_3 . H_3 consumes more energy than H_2 in October, as the temperature in this month is cold (sometimes below 20°C in the dataset). A low temperature, which is not in the predefined interval, leads to frequent sensor activations despite the lack of significant temperature variation. In July and August, energy consumption is nearly the same for H_2 and H_3 , as high temperature variation induces frequent sensor activations.

H_1 provides periodic collection, but may fail to detect sudden changes in hive. H_2 offers the best responsiveness by

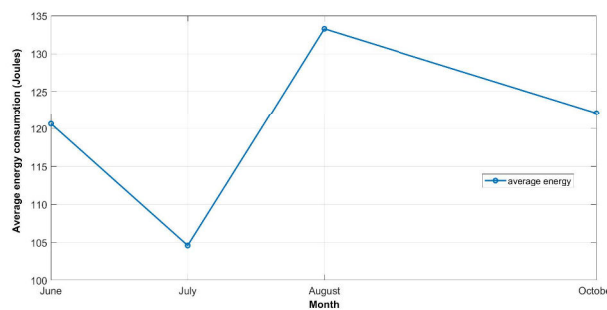


Fig. 4. Energy consumption per month using heuristic 3

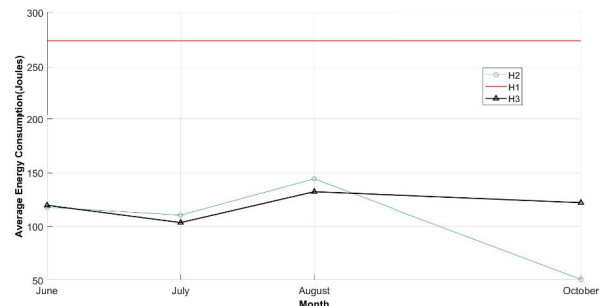


Fig. 5. Comparison of results induced by different data sampling heuristics

activating sensors immediately when temperature thresholds are exceeded. H_3 provides a balanced approach, ensuring data collection of critical events without excessive energy use.

To conclude, H_1 is useful for stable environments where constant data collection is needed. H_2 is ideal for dynamic conditions, ensuring detection of sudden hive changes. H_3 is best for remote hives, balancing energy savings and effective monitoring. The selection of the suitable heuristic to be implemented is primarily dependent on the beekeeper requirements.

VII. CONCLUSION

In this paper, we proposed an energy-efficient IoT-based beehive system. This system combines the use of energy-efficient communication protocols to ensure reliable data transfer, and data sampling and transmission heuristics designed to reduce energy consumption while maintaining consistent monitoring. Through simulation-based performance evaluation, the system demonstrated its ability to achieve significant energy savings without compromising reliability. These innovations make the system particularly effective for remote and rural areas, where the frequent battery replacement is impractical. Further improvements can enhance the proposed solution. Machine learning techniques may refine data analysis, facilitating early issue detection. Field testing and validation in real-world conditions would provide critical insights into system performance. Additionally, developing sophisticated heuristics could further optimize data collection and transmission strategies.

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